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B. E. (Third Semester) Examination, Nov. Dec. 2016

(New Course)

(Mechnical Engineering Branch)

MATHEMATICS-III

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt part (a) which is compulsory and any two parts from (b), (c) and (d) of each question. All questions carry equal marks.

Unit- I

1. (a) Define Fourier series. 2

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(b) Obtain a fourier series for the function

$f(x)=|x|, -\pi < x < \pi$.

Hence deduce that :

$$\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots=\frac{\pi^2}{8}.$$ 7

(c) Find the half range cosine series for the function

$f(x)=x^2$ in the range $0 \le x \le \pi$. 7

(d) The following table given the variations of periodic current over a period.

t (Second)	A (Amp)
0	1·98
T/6	1·30
T/3	1·05
T/2	1·30
2T/3	–0·88
5T/6	–0·25
T	1·98

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Show that there is a direct current part of 0·75 Ampere in the variable current and obtain the amplitude of the first harmonic. 7

Unit- II

2. (a) Find $L\{1\}$. 2

(b) Find the Laplace transform of $\frac{\sin at}{t}$. Does the transform of $\frac{\cos at}{t}$ exist. 7

(c) Evaluate :

$$L^{-1}\left\{\frac{S+2}{S^2-4S+13}\right\}$$ 7

(d) Use convolution theorem, to find

$$L^{-1}\left\{\frac{S}{\left(S^2+a^2\right)^2}\right\}$$ 7

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Unit- III

3. (a) Form the P.D.E. from $z=f\left(x^2-y^2\right)$ by eliminating the arbitrary function. 2

(b) Solve the equation :

$$p \tan x + q \tan y = \tan z.$$ 7

(c) Solve :

$$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial x \partial y}-6\frac{\partial^2 z}{\partial y^2}=\cos(2x+y)$$ 7

(d) Using the method of separation of variables, Solve

$$\frac{\partial u}{\partial x}=2\frac{\partial u}{\partial t}+u,$$

where $u(x,0)=6e^{-3x}$ 7

Unit- IV

4. (a) Write polar form of cauchy-Riemann equations. 2

(b) Determine p such that the function

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$f(z)=\frac{1}{2}\log\left(x^2+y^2\right)+i\tan^{-1}\left(\frac{px}{y}\right)$ be an analytic function. 7

(c) Evaluate using Cauchy integral formula

$$\int_C \frac{e^{2z}}{(z-1)(z-2)}$$

where C is the circle $|z|=3$. 7

(d) Apply the calculus of residues to prove that :

$$\int_0^{2\pi}\frac{d\theta}{1-2p\sin\theta+p^2}=\frac{2\pi}{1-p^2},(0<\phi<1).$$ 7

Unit- V

5. (a) Define variance of a distribution. 2

(b) The p.d.f. of a variate X is :

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X	$p(X)$
0	k
1	3k
2	5k
3	7k
4	9k
5	11k
6	13K

(i) Find :

$$P(X<4), P(X\geq 5), P(3<X\leq 6)$$

(ii) What will be the minimum value of k so that :

$P(X\leq 2)>0{\cdot}3$? 7

(c) Find the moment generating function of the exponential distribution :

$$f(x)=\frac{1}{C}e^{-x/c},\ 0<x<\infty, c>0$$

Hence find its mean and S.D. 7

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(d) Fit a Poisson distribution to the following : 7

x	f
0	16
1	38
2	22
3	9
4	1

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